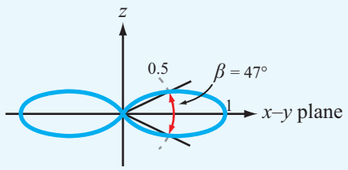
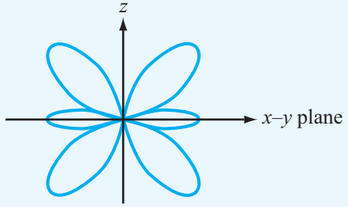



(a) $l = \lambda/2$




(b) $l = \lambda$




(c) $l = 3\lambda/2$

**Figure 9-17** Radiation patterns of dipoles with lengths of $\lambda/2$, $\lambda$, and $3\lambda/2$.

**Module 9.2 Linear Dipole Antenna** For a linear antenna of any specified length (in units of $\lambda$), this module displays the current distribution along the antenna and the far-field radiation patterns in the horizontal and elevation planes. It also calculates the total power radiated by the antenna, the radiation resistance, and the antenna directivity.

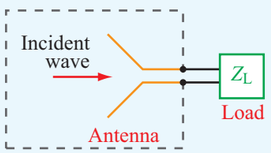
## 9-5 Effective Area of a Receiving Antenna

So far, antennas have been treated as directional radiators of energy. Now, we examine the reverse process, namely how a receiving antenna extracts energy from an incident wave and delivers it to a load. The ability of an antenna to capture energy from an incident wave of power density $S_i$ (W/m$^2$) and convert it into an ***intercepted power*** $P_{int}$ (W) for delivery to a matched load is characterized by the ***effective area*** $A_e$:

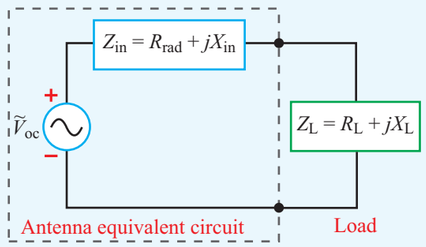
$$A_e = \frac{P_{int}}{S_i} \quad (m^2). \tag{9.57}$$

Other commonly used names for $A_e$ include ***effective aperture*** and ***receiving cross section***. The antenna receiving process may be modeled in terms of a Thévenin equivalent circuit (**Fig. 9-18**) consisting of a voltage $\widetilde{V}_{oc}$ in series with the antenna input impedance $Z_{in}$. Here, $\widetilde{V}_{oc}$ is the open-circuit voltage induced by the incident wave at the antenna terminals, and $Z_L$ is the impedance of the load connected to the antenna (representing a receiver or some other circuit). In general, both $Z_{in}$ and $Z_L$ are complex:

$$Z_{in} = R_{rad} + jX_{in}, \tag{9.58a}$$




(a) Receiving antenna

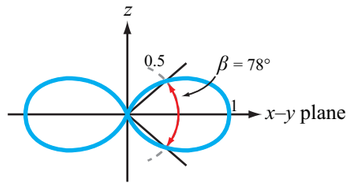



(b) Equivalent circuit

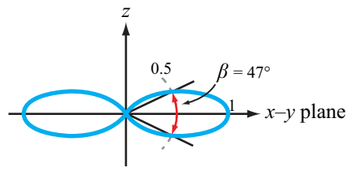
**Figure 9-18** Receiving antenna represented by an equivalent circuit.



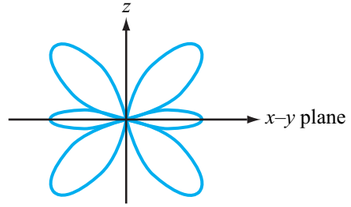





(a) $l = \lambda/2$




(b) $l = \lambda$




(c) $l = 3\lambda/2$

图 9-17 长度为 λ/2、λ 和 3λ/2 的偶极子的辐射方向图。

模块 9.2 线性偶极子天线 对于任何指定长度（以 λ 为单位）的线性天线，该模块显示沿天线的电流分布以及水平面和仰角面中的远场辐射方向图。它还计算天线辐射的总功率、辐射电阻和天线方向性。

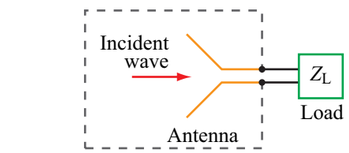
## 9-5 接收天线的有效面积

到目前为止，天线已被视为能量的定向辐射器。现在，我们研究相反的过程，即接收天线如何从入射波中提取能量并将其传递给负载。天线从功率密度为 $S_i$ (W/m$^2$) 的入射波捕获能量并将其转换为截获功率 $P_{int}$ (W) 以传送到匹配负载的能力由有效面积 $A_e$ 来表征：

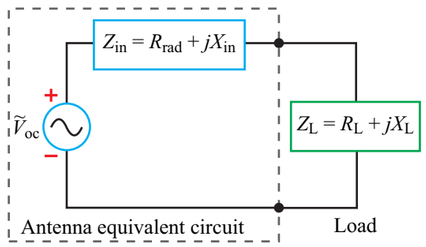
$$A_e = \frac{P_{int}}{S_i} \quad (m^2). \tag{9.57}$$

$A_e$ 的其他常用名称包括有效孔径和接收横截面。天线接收过程可以根据Thévenin等效电路（图9-18）进行建模，该电路由与天线输入阻抗$Z_{in}$串联的电压$\widetilde{V}_{oc}$组成。这里，$\widetilde{V}_{oc}$是天线端子处的入射波感应的开路电压，$Z_L$是连接到天线的负载的阻抗（代表接收器或其他电路）。一般来说，$Z_{in}$ 和 $Z_L$ 都是复数：

$$Z_{in} = R_{rad} + jX_{in}, \tag{9.58a}$$



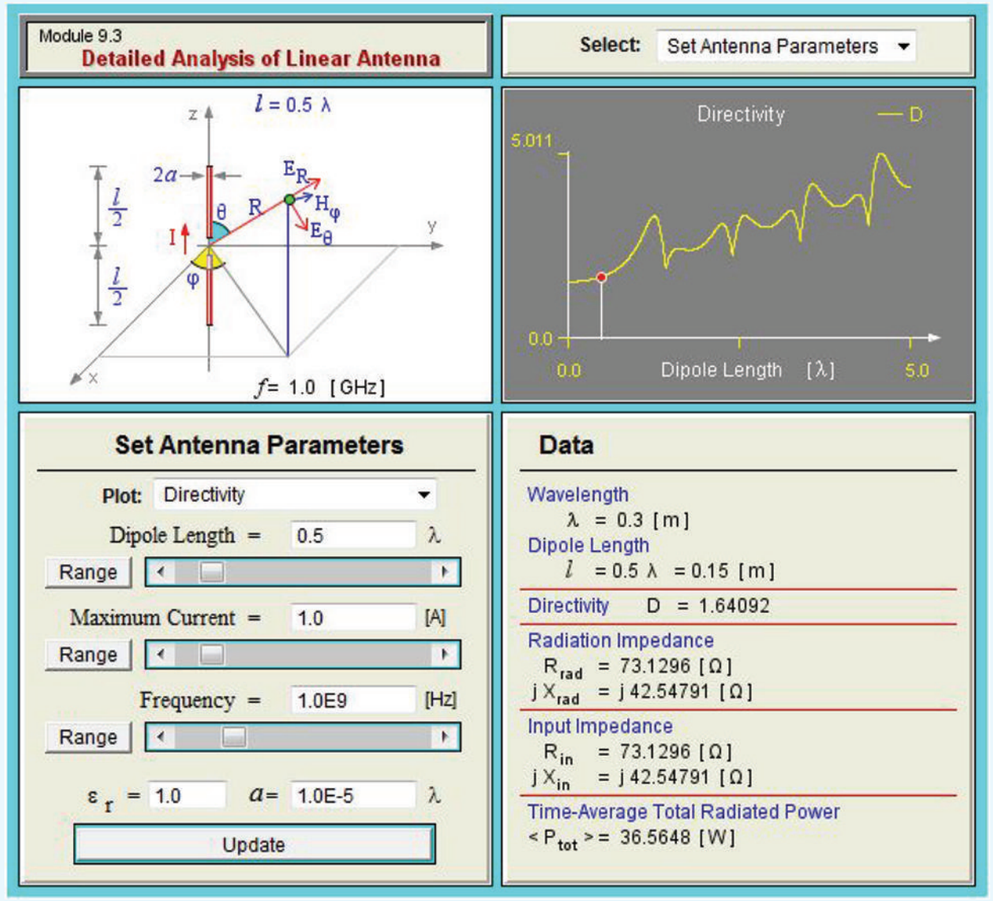

(a) Receiving antenna




(b) Equivalent circuit

图 9-18 用等效电路表示的接收天线。

**Module 9.3 Detailed Analysis of Linear Antenna** This module complements Module 9.2 by offering extensive information about the specified linear antenna, including its directivity and plots of its current and field distributions.




$$Z_L = R_L + jX_L, \tag{9.58b}$$

where $R_{rad}$ denotes the radiation resistance of the antenna (assuming $R_{loss} \ll R_{rad}$). To maximize power transfer to the load, the load impedance must be chosen so that either $Z_L = Z_{in}^*$, or equivalently, $R_L = R_{rad}$ and $X_L = -X_{in}$. In that case, the circuit reduces to a source $\widetilde{V}_{oc}$ that is connected across a resistance equal to $2R_{rad}$. Since $V_{oc}$ is a sinusoidal voltage phasor, the time-average power delivered to the load is

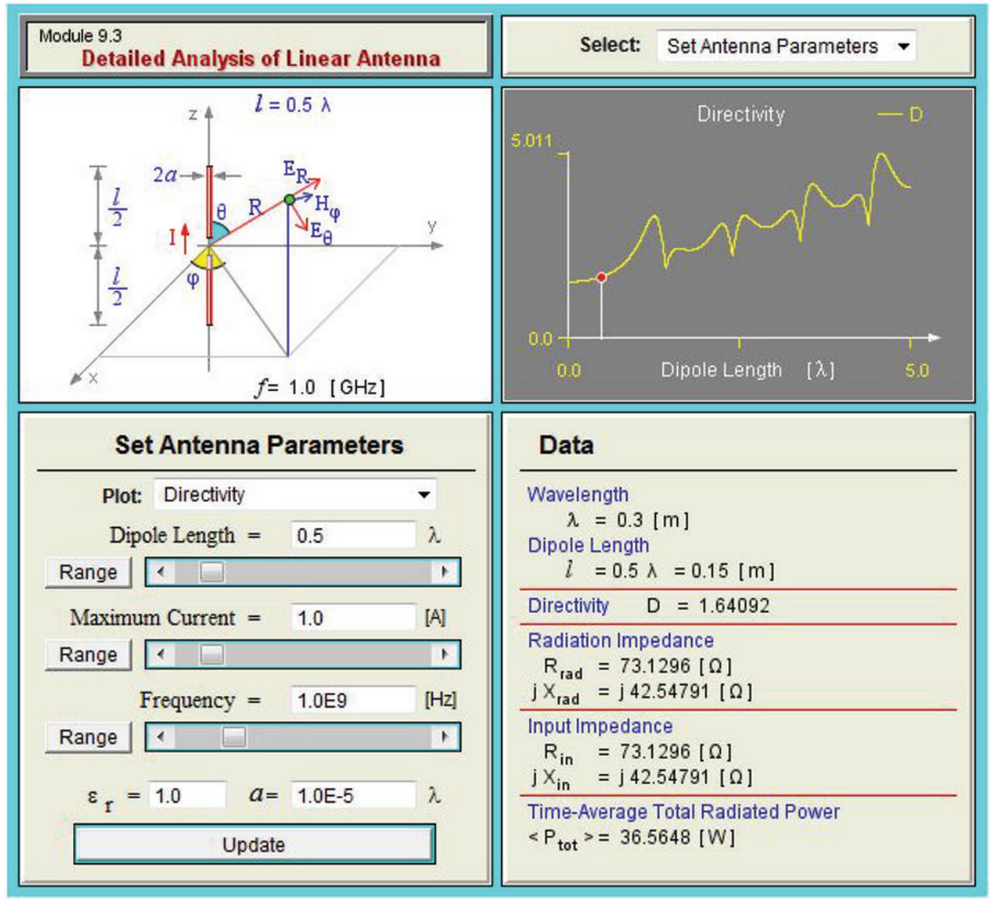
$$P_L = \frac{1}{2}|\widetilde{I}_L|^2 R_{rad} = \frac{1}{2}\left[\frac{|\widetilde{V}_{oc}|}{2R_{rad}}\right]^2 R_{rad} = \frac{|\widetilde{V}_{oc}|^2}{8R_{rad}}, \tag{9.59}$$

where $\widetilde{I}_L = \widetilde{V}_{oc}/(2R_{rad})$ is the phasor current flowing through the circuit. Since the antenna is lossless, all of the intercepted



402 CHAPTER 9 RADIATION AND ANTENNAS

模块 9.3 线性天线的详细分析 该模块通过提供有关指定线性天线的广泛信息（包括其方向性及其电流和场分布图）来补充模块 9.2。




$$Z_L = R_L + jX_L, \tag{9.58b}$$

其中$R_{rad}$表示天线的辐射电阻（假设$R_{loss} \ll R_{rad}$）。为了最大限度地提高向负载传输的功率，必须选择负载阻抗，以便 $Z_L = Z_{in}^*$ 或等效的 $R_L = R_{rad}$ 和 $X_L = -X_{in}$。在这种情况下，电路简化为源极 $\widetilde{V}_{oc}$，其跨接等于 $2R_{rad}$ 的电阻。由于 $\widetilde{V}_{oc}$ 是正弦电压

相量，输送到负载的时间平均功率为

$$P_L = \frac{1}{2}|\widetilde{I}_L|^2 R_{rad} = \frac{1}{2}\left[\frac{|\widetilde{V}_{oc}|}{2R_{rad}}\right]^2 R_{rad} = \frac{|\widetilde{V}_{oc}|^2}{8R_{rad}}, \tag{9.59}$$

其中$\widetilde{I}_L = \widetilde{V}_{oc}/(2R_{rad})$是流过电路的相量电流。由于天线是无损的，所有截获的

power $P_{int}$ ends up in the load resistance $R_L$. Hence,

$$P_{int} = P_L = \frac{|\widetilde{V}_{oc}|^2}{8R_{rad}}. \tag{9.60}$$

For an incident wave with electric field $\widetilde{E}_i$ parallel to the antenna polarization direction, the power density carried by the wave is

$$S_i = \frac{|\widetilde{E}_i|^2}{2\eta_0} = \frac{|\widetilde{E}_i|^2}{240\pi}. \tag{9.61}$$

The ratio of the results provided by Eqs. (9.60) and (9.61) gives

$$A_e = \frac{P_{int}}{S_i} = \frac{30\pi|\widetilde{V}_{oc}|^2}{R_{rad}|\widetilde{E}_i|^2}. \tag{9.62}$$

The open-circuit voltage $\widetilde{V}_{oc}$ induced in the receiving antenna is due to the incident field $\widetilde{E}_i$, but the relation between them depends on the specific antenna under consideration. By way of illustration, let us consider the case of the short-dipole antenna of Section 9-1. Because the length $l$ of the short dipole is small compared with $\lambda$, the current induced by the incident field is uniform across its length, and the open-circuit voltage is simply $\widetilde{V}_{oc} = \widetilde{E}_i l$. Noting that $R_{rad} = 80\pi^2(l/\lambda)^2$ for the short dipole (see Eq. (9.38)) and using $\widetilde{V}_{oc} = \widetilde{E}_i l$, Eq. (9.62) simplifies to

$$A_e = \frac{3\lambda^2}{8\pi} \quad (m^2). \quad \textbf{(short dipole)} \tag{9.63}$$

In Example 9-2 it was shown that for the Hertzian dipole the directivity $D = 1.5$. In terms of $D$, Eq. (9.63) can be rewritten in the form

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (m^2). \quad \textbf{(any antenna)} \tag{9.64}$$

► Despite the fact that the relation between $A_e$ and $D$ given by Eq. (9.64) was derived for a Hertzian dipole, it can be shown that it is also valid *for any antenna* under matched-impedance conditions. ◄

**Exercise 9-6:** The effective area of an antenna is 9 $m^2$. What is its directivity in decibels at 3 GHz?

**Answer:** $D = 40.53$ dB. (See EM.)

**Exercise 9-7:** At 100 MHz, the pattern solid angle of an antenna is 1.3 sr. Find (a) the antenna directivity $D$ and (b) its effective area $A_e$.

**Answer:** (a) $D = 9.67$, (b) $A_e = 6.92\ m^2$. (See EM.)

## 9-6 Friis Transmission Formula

The two antennas shown in **Fig. 9-19** are part of a free-space communication link with the separation between the antennas, $R$, being large enough for each to be in the far-field region of the other. The transmitting and receiving antennas have effective areas $A_t$ and $A_r$ and radiation efficiencies $\xi_t$ and $\xi_r$, respectively. Our objective is to find a relationship between $P_t$, which is the power supplied to the transmitting antenna, and $P_{rec}$, which is the power delivered to the receiver. As always, we assume that both antennas are impedance-matched to their respective transmission lines. Initially, we consider the case where the two antennas are oriented so that the peak of the radiation pattern of each antenna points in the direction of the other.

We start by treating the transmitting antenna as a lossless isotropic radiator. The power density incident upon the receiving antenna at a distance $R$ from an isotropic transmitting antenna is simply equal to the transmitter power $P_t$ divided by the surface area of a sphere of radius $R$:

$$S_{iso} = \frac{P_t}{4\pi R^2}. \tag{9.65}$$

The real transmitting antenna is neither lossless nor isotropic. Hence, the power density $S_r$ due to the real antenna is

$$S_r = G_t S_{iso} = \xi_t D_t S_{iso} = \frac{\xi_t D_t P_t}{4\pi R^2}. \tag{9.66}$$

Through the gain $G_t = \xi_t D_t$, $\xi_t$ accounts for the fact that only part of the power $P_t$ supplied to the antenna is radiated out into space, and $D_t$ accounts for the directivity of the transmitting

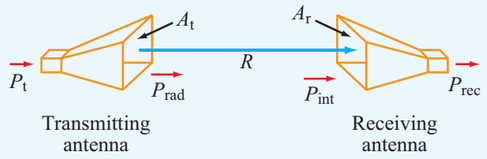



**Figure 9-19** Transmitter–receiver configuration.





功率 $P_{int}$ 最终变为负载电阻 $R_L$。因此，

$$P_{int} = P_L = \frac{|\widetilde{V}_{oc}|^2}{8R_{rad}}. \tag{9.60}$$

对于电场 $\widetilde{E}_i$ 平行于天线极化方向的入射波，波所携带的功率密度为

$$S_i = \frac{|\widetilde{E}_i|^2}{2\eta_0} = \frac{|\widetilde{E}_i|^2}{240\pi}. \tag{9.61}$$

等式提供的结果的比率。(9.60) 和 (9.61) 给出

$$A_e = \frac{P_{int}}{S_i} = \frac{30\pi|\widetilde{V}_{oc}|^2}{R_{rad}|\widetilde{E}_i|^2}. \tag{9.62}$$

接收天线中感应的开路电压 $\widetilde{V}_{oc}$ 是由入射场 $\widetilde{E}_i$ 引起的，但它们之间的关系取决于所考虑的特定天线。作为说明，让我们考虑第 9-1 节的短偶极子天线的情况。由于短偶极子的长度 l 与 λ 相比较小，因此入射场感应的电流在其长度上是均匀的，开路电压仅为 $\widetilde{V}_{oc} = \widetilde{E}_i l$。注意到 $R_{rad} = 80\pi^2(l/\lambda)^2$ 对于短偶极子（参见方程（9.38））并使用 $\widetilde{V}_{oc} = \widetilde{E}_i l$，方程(9.62) 简化为

$$A_e = \frac{3\lambda^2}{8\pi} \quad (m^2). \quad \textbf{(short dipole)} \tag{9.63}$$

在实施例9-2中表明，对于赫兹偶极子，方向性D=为1.5。就 D 而言，方程。(9.63) 可以重写为以下形式

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (m^2). \quad \textbf{(any antenna)} \tag{9.64}$$

► 尽管事实上 $A_e$ 和 D 之间的关系由方程给出。式（9.64）是针对赫兹偶极子导出的，可以证明它对于匹配阻抗条件下的任何天线也有效。◄

练习9-6：天线的有效面积为9 $m^2$。 3 GHz 时的方向性（以分贝为单位）是多少？

**Answer:** $D = 40.53$ dB. (See EM.)

练习 9-7：在 100 MHz 时，天线的方向图立体角为 1.3 sr。求 (a) 天线方向性 D 和 (b) 其有效面积 $A_e$。

**Answer:** (a) $D = 9.67$, (b) $A_e = 6.92\ m^2$. (See EM.)

## 9-6 弗里斯传输公式

图 9-19 中所示的两个天线是自由空间通信链路的一部分，天线之间的间隔 R 足够大，足以使每个天线位于另一个天线的远场区域中。发射和接收天线的有效面积分别为 $A_t$ 和 $A_r$，辐射效率分别为 $\xi_t$ 和 $\xi_r$。我们的目标是找到 $P_t$（提供给发射天线的功率）和 $P_{rec}$（提供给接收器的功率）之间的关系。一如既往，我们假设两个天线都与其各自的传输线阻抗匹配。最初，我们考虑两个天线定向的情况，使得每个天线的辐射方向图的峰值指向另一个天线的方向。

我们首先将发射天线视为无损各向同性辐射器。距各向同性发射天线距离 R 处入射到接收天线的功率密度简单地等于发射器功率 $P_t$ 除以半径为 R 的球体表面积：

$$S_{iso} = \frac{P_t}{4\pi R^2}. \tag{9.65}$$

真正的发射天线既不是无损的，也不是各向同性的。因此，实际天线的功率密度 $S_r$ 为

$$S_r = G_t S_{iso} = \xi_t D_t S_{iso} = \frac{\xi_t D_t P_t}{4\pi R^2}. \tag{9.66}$$

通过增益 $G_t = \xi_t D_t$，$\xi_t$ 说明了只有部分提供给天线的功率 $P_t$ 辐射到空间中的事实，而 $D_t$ 则说明了发射的方向性

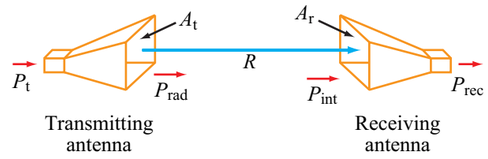



图 9-19 发射器-接收器配置。

antenna (in the direction of the receiving antenna). Moreover, by Eq. (9.64), $D_t$ is related to $A_t$ by $D_t = 4\pi A_t/\lambda^2$. Hence, Eq. (9.66) becomes

$$S_r = \frac{\xi_t A_t P_t}{\lambda^2 R^2}. \tag{9.67}$$

On the receiving-antenna side, the power intercepted by the receiving antenna is equal to the product of the incident power density $S_r$ and the effective area $A_r$:

$$P_{int} = S_r A_r = \frac{\xi_t A_t A_r P_t}{\lambda^2 R^2}. \tag{9.68}$$

The power delivered to the receiver, $P_{rec}$, is equal to the intercepted power $P_{int}$ multiplied by the radiation efficiency of the receiving antenna, $\xi_r$. Hence, $P_{rec} = \xi_r P_{int}$, which leads to the result

$$\frac{P_{rec}}{P_t} = \frac{\xi_t \xi_r A_t A_r}{\lambda^2 R^2} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{9.69}$$

▶ This relation is known as the ***Friis transmission formula***, and $P_{rec}/P_t$ is called the ***power transfer ratio***. ◀

If the two antennas are not oriented in the direction of maximum power transfer, Eq. (9.69) assumes the general form

$$\frac{P_{rec}}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 F_t(\theta_t, \phi_t)\, F_r(\theta_r, \phi_r), \tag{9.70}$$

where $F_t(\theta_t, \phi_t)$ is the normalized radiation intensity of the transmitting antenna at angles $(\theta_t, \phi_t)$ corresponding to the direction of the receiving antenna (as seen by the antenna pattern of the transmitting antenna), and a similar definition applies to $F_r(\theta_r, \phi_r)$ for the receiving antenna.

## Example 9-4: Satellite Communication System

A 6 GHz direct-broadcast TV satellite system transmits 100 W through a 2 m diameter parabolic dish antenna from a distance of approximately 40,000 km above Earth's surface. Each TV channel occupies a bandwidth of 5 MHz. Due to electromagnetic noise picked up by the antenna as well as noise generated by the receiver electronics, a home TV receiver has a noise level given by

$$P_n = KT_{sys}B \quad \text{(W)}, \tag{9.71}$$

where $T_{sys}$ [measured in kelvins (K)] is a figure of merit called the ***system noise temperature*** that characterizes the noise performance of the receiver–antenna combination, $K$ is Boltzmann's constant $[1.38 \times 10^{-23}$ (J/K)], and $B$ is the receiver bandwidth in Hz.

The ***signal-to-noise ratio*** $S_n$ (which should not be confused with the power density $S$) is defined as the ratio of $P_{rec}$ to $P_n$:

$$S_n = P_{rec}/P_n \quad \text{(dimensionless)}. \tag{9.72}$$

For a receiver with $T_{sys} = 580$ K, what minimum diameter of a parabolic dish receiving antenna is required for high-quality TV reception with $S_n = 40$ dB? The satellite and ground receiving antennas may be assumed lossless, and their effective areas may be assumed equal to their physical apertures.

**Solution:** The following quantities are given:

$$P_t = 100 \text{ W}, \quad f = 6 \text{ GHz} = 6 \times 10^9 \text{ Hz}, \quad S_n = 10^4,$$
$$\text{transmit antenna diameter } d_t = 2 \text{ m},$$
$$T_{sys} = 580 \text{ K}, \quad R = 40{,}000 \text{ km} = 4 \times 10^7 \text{ m},$$
$$B = 5 \text{ MHz} = 5 \times 10^6 \text{ Hz}.$$

The wavelength $\lambda = c/f = 5 \times 10^{-2}$ m, and the area of the transmitting satellite antenna is $A_t = (\pi d_t^2/4) = \pi$ (m$^2$). From Eq. (9.71), the receiver noise power is

$$P_n = KT_{sys}B = 1.38 \times 10^{-23} \times 580 \times 5 \times 10^6 = 4 \times 10^{-14} \text{ W}.$$

Using Eq. (9.69) with $\xi_t = \xi_r = 1$,

$$P_{rec} = \frac{P_t A_t A_r}{\lambda^2 R^2} = \frac{100\pi A_r}{(5 \times 10^{-2})^2 (4 \times 10^7)^2} = 7.85 \times 10^{-11} A_r.$$

The area of the receiving antenna, $A_r$, can now be determined by equating the ratio $P_{rec}/P_n$ to $S_n = 10^4$:

$$10^4 = \frac{7.85 \times 10^{-11} A_r}{4 \times 10^{-14}},$$

which yields the value $A_r = 5.1$ m$^2$. The required minimum diameter is $d_r = \sqrt{4A_r/\pi} = 2.55$ m.

**Exercise 9-8:** If the operating frequency of the communication system described in Example 9-4 is doubled to 12 GHz, what would then be the minimum required diameter of a home receiving TV antenna?

**Answer:** $d_r = 1.27$ m. (See EM.)

**Exercise 9-9:** A 3 GHz microwave link consists of two identical antennas each with a gain of 30 dB. Determine the received power given that the transmitter output power is 1 kW and the two antennas are 10 km apart.

**Answer:** $P_{rec} = 6.33 \times 10^{-4}$ W. (See EM.)



404 CHAPTER 9 RADIATION AND ANTENNAS

天线（接收天线方向）。此外，通过等式。(9.64)，$D_t$ 通过 $D_t = 4\pi A_t/\lambda^2$ 与 $A_t$ 相关。因此，等式。(9.66) 变为

$$S_r = \frac{\xi_t A_t P_t}{\lambda^2 R^2}. \tag{9.67}$$

在接收天线侧，接收天线截获的功率等于入射功率密度$S_r$与有效面积$A_r$的乘积：

$$P_{int} = S_r A_r = \frac{\xi_t A_t A_r P_t}{\lambda^2 R^2}. \tag{9.68}$$

传送到接收器的功率 $P_{rec}$ 等于拦截功率 $P_{int}$ 乘以接收天线的辐射效率 $\xi_r$。因此，$P_{rec} = \xi_r P_{int}$，导致结果

$$\frac{P_{rec}}{P_t} = \frac{\xi_t \xi_r A_t A_r}{\lambda^2 R^2} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{9.69}$$

▶ 这种关系称为弗里斯传输公式，$P_{rec}/P_t$称为功率传输比。◀

如果两个天线没有朝向最大功率传输的方向，则方程：(9.69) 呈现一般形式

$$\frac{P_{rec}}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 F_t(\theta_t, \phi_t)\, F_r(\theta_r, \phi_r), \tag{9.70}$$

其中 $F_t(\theta_t、\phi_t)$ 是发射天线在与接收天线方向（如发射天线的天线方向图所示）相对应的角度（$\theta_t、\phi_t$）处的归一化辐射强度，类似的定义适用于接收天线的 $F_r(\theta_r、\phi_r)$。

## 例9-4：卫星通信系统

6 GHz 直播电视卫星系统通过直径 2 m 的抛物面碟形天线从距地球表面约 40,000 公里的距离传输 100 W 功率。每个电视频道占用 5 MHz 的带宽。由于天线接收到的电磁噪声以及接收器电子设备产生的噪声，家庭电视接收器的噪声水平为：

$$P_n = KT_{sys}B \quad \text{(W)}, \tag{9.71}$$

其中 $T_{sys}$ [以开尔文 (K) 为单位测量] 是一个品质因数，称为系统噪声温度，它表征了噪声接收器-天线组合的性能，K 是玻尔兹曼常数 $[1.38 \times 10^{-23}$ (J/K)]，B 是接收器带宽（以 Hz 为单位）。

信噪比 $S_n$（不应与功率密度 S）混淆，其定义为 $P_{rec}$ 与 $P_n$ 之比：

$$S_n = P_{rec}/P_n \quad \text{(dimensionless)}. \tag{9.72}$$

对于 $T_{sys} = 580$ K 的接收器，要实现 $S_n = 40$ dB 的高质量电视接收，抛物面碟形接收天线的最小直径需要多少？可以假设卫星和地面接收天线是无损的，并且可以假设它们的有效面积等于它们的物理孔径。

解：给出下列数量：

$$P_t = 100 \text{ W}, \quad f = 6 \text{ GHz} = 6 \times 10^9 \text{ Hz}, \quad S_n = 10^4,$$
$$\text{transmit antenna diameter } d_t = 2 \text{ m},$$
$$T_{sys} = 580 \text{ K}, \quad R = 40{,}000 \text{ km} = 4 \times 10^7 \text{ m},$$
$$B = 5 \text{ MHz} = 5 \times 10^6 \text{ Hz}.$$

波长为$\lambda = c/f = 5 \times 10^{-2}$m，发射卫星天线面积为$A_t = (\pi d_t^2/4) = \pi$ (m$^2$)。从方程。(9.71)，接收器噪声功率为

$$P_n = KT_{sys}B = 1.38 \times 10^{-23} \times 580 \times 5 \times 10^6 = 4 \times 10^{-14} \text{ W}.$$

使用方程式(9.69) 与 $\xi_t = \xi_r = 1$,

$$P_{rec} = \frac{P_t A_t A_r}{\lambda^2 R^2} = \frac{100\pi A_r}{(5 \times 10^{-2})^2 (4 \times 10^7)^2} = 7.85 \times 10^{-11} A_r.$$

现在可以通过将 $P_{rec}/P_n$ 与 $S_n = 10^4$ 的比率相等来确定接收天线的面积 $A_r$：

$$10^4 = \frac{7.85 \times 10^{-11} A_r}{4 \times 10^{-14}},$$

得出值 $A_r = 5.1$ m$^2$。所需的最小直径为 $d_r = \sqrt{4A_r/\pi} = 2.55$ m。

练习9-8：如果例9-4中描述的通信系统的工作频率加倍到12 GHz，那么家庭接收电视天线所需的最小直径是多少？

**Answer:** $d_r = 1.27$ m. (See EM.)

练习 9-9：3 GHz 微波链路由两个相同的天线组成，每个天线的增益均为 30 dB。假定发射机输出功率为 1 kW，两个天线相距 10 km，确定接收功率。

**Answer:** $P_{rec} = 6.33 \times 10^{-4}$ W. (See EM.)

## Technology Brief 17: Health Risks of EM Fields

Can the use of cell phones cause cancer? Does exposure to the electromagnetic fields (EMFs) associated with power lines pose health risks to humans? Are we endangered by EMFs generated by home appliances, telephones, electrical wiring, and the myriad of electronic gadgets we use every day (**Fig. TF17-1**)? Despite reports in some of the popular media alleging a causative relationship between low-level EMFs and many diseases, according to reports issude by governmental and professional boards in the U.S. and Europe, the answer is given here.

► NO, we are not at risk, so long as manufacturers adhere to the approved governmental standards for ***maximum permissible exposure*** (***MPE***) levels. With regard to cell phones, the official reports caution that their conclusions are limited to phone use of less than 15 years, since data for longer use is not yet available. ◄



**Figure TF17-1** Electromagnetic fields are emitted by power lines, cell phones, TV towers, and many other electronic circuits and devices.

### Physiological Effects of EMFs

The energy carried by a photon with an EM frequency $f$ is given by $E = hf$, where $h$ is Planck's constant. The mode of interaction between a photon passing through a material and the material's atoms or molecules is very much dependent on $f$. If $f$ is greater than about $10^{15}$ Hz (which falls in the ultraviolet (UV) band of the EM spectrum), the photon's energy is sufficient to free an electron and remove it completely, thereby ionizing the affected atom or molecule. Consequently, the energy carried by such EM waves is called ***ionizing radiation***, in contrast with ***non-ionizing radiation*** (**Fig. TF17-2**), where photons may be able to cause an electron to move to a higher energy level but not eject it from its host atom or molecule.

Assessing health risks associated with exposure to EMFs is complicated by the number of variables involved, including: (1) the frequency $f$, (2) the intensities of the electric and magnetic fields, (3) the exposure duration—whether continuous or discontinuous, pulsed or uniform—and (4) the specific part of the body that is getting exposed. We know that intense laser illumination can cause corneal burn, high-level X-rays can damage living tissue and cause cancer, and in fact, any form of EM energy can be dangerous if the exposure level and/or duration exceed certain safety limits. Governmental and professional safety boards are tasked with establishing maximum permissible exposure (MPE) levels that protect human beings against adverse health effects associated with EMFs. In the U.S., the relevant standards are IEEE Std C95.6 (dated 2002), which addresses EM fields in the 1 Hz to 3 kHz range, and IEEE Std 95.1 (dated 2005), which deals with the frequency range from 3 kHz to 300 GHz. On the European side of the Atlantic, responsibility for establishing MPE levels resides with the Scientific Committee on Emerging and Newly Identified Health Risks (SCENIHR) of the European Commission.

► At frequencies below 100 kHz, the goal is to minimize adverse effects of exposure to electric fields that can cause ***electrostimulation*** of nerve and muscle cells. Above 5 MHz, the main concern is excessive tissue heating, and in the transition region of 100 kHz to 5 MHz, safety standards are designed to protect against both electrostimulation and excessive heating. ◄





## 技术简介 17：电磁场的健康风险

使用手机会致癌吗？暴露于与电源线相关的电磁场 (EMF) 会对人类造成健康风险吗？我们是否受到家用电器、电话、电线以及我们每天使用的无数电子产品产生的电磁场的威胁（图TF17-1）？尽管一些流行媒体报道称低水平电磁场与许多疾病之间存在因果关系，但根据美国和欧洲政府和专业委员会发布的报告，这里给出了答案。

► 不，只要制造商遵守政府批准的最大允许暴露 (MPE) 水平标准，我们就不会面临风险。关于手机，官方报告警告说，他们的结论仅限于手机使用时间少于 15 年的情况，因为尚无更长时间使用的数据。 ◄



图 TF17-1 电磁场由电源线、手机、电视塔以及许多其他电子电路和设备发射。

### 电磁场的生理效应

电磁频率为 f 的光子携带的能量由 E = h f 给出，其中 h 是普朗克常数。穿过材料的光子与材料的原子或分子之间的相互作用模式很大程度上取决于 f 。如果 f 大于约 $10^{15}$ Hz（落在电磁光谱的紫外 (UV) 波段），则光子的能量足以释放电子并将其完全移除，从而电离受影响的原子或分子。因此，这种电磁波携带的能量称为电离辐射，与非电离辐射相反（图TF17-2），其中光子可能使电子移动到更高的能级，但不会将其从其主体原子或分子中喷射出来。

评估与暴露于电磁场相关的健康风险因涉及的变量数量而变得复杂，包括：(1) 频率 f、(2) 电场和磁场的强度、(3) 暴露持续时间（无论是连续还是不连续、脉冲还是均匀）以及 (4) 暴露的身体特定部位。我们知道，强烈的激光照射会导致角膜烧伤，高强度 X 射线会损伤活体组织并导致癌症，事实上，如果暴露水平和/或持续时间超过一定的安全限度，任何形式的电磁能量都可能是危险的。政府和专业安全委员会的任务是确定最大允许暴露 (MPE) 水平，以保护人类免受与电磁场相关的不良健康影响。在美国，相关标准包括 IEEE Std C95.6（2002 年）和 IEEE Std 95.1（2005 年），前者涉及 1 Hz 至 3 kHz 范围内的电磁场，后者涉及 3 kHz 至 300 GHz 频率范围。在大西洋的欧洲一侧，欧盟委员会新兴和新发现的健康风险科学委员会 (SCENIHR) 负责制定 MPE 水平。

► 在低于 100 kHz 的频率下，目标是最大限度地减少暴露于电场中可能导致神经和肌肉细胞电刺激的不利影响。高于 5 MHz 时，主要问题是组织过度加热，而在 100 kHz 至 5 MHz 的过渡区域，安全标准旨在防止电刺激和过度加热。 ◄